

High-Frequency Market Impact of Monetary Policy Operations

A Simulation-Based Causal Inference Study

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Abstract

This study develops a synthetic high-frequency market environment to evaluate the causal impact of monetary policy operations on returns, volatility, and liquidity. Using Double Machine Learning and Causal Forest estimators, the analysis quantifies both global and heterogeneous treatment effects. The framework provides a transparent, reproducible foundation for assessing policy execution strategies in modern financial markets.

Executive Summary

Monetary policy operations influence financial markets at high frequency, yet their causal effects are difficult to isolate in real-world data due to confounding, noise, and limited transparency. This study introduces a fully synthetic but realistic high-frequency market simulator that embeds microstructure dynamics, liquidity regimes, and heterogeneous policy impacts. The simulator provides a controlled environment where the **true causal effect is known**, enabling rigorous evaluation of modern causal-inference methods.

Policy operations are modeled as temporary shocks to drift and volatility, with effect sizes that depend on market conditions — particularly trading volume. This creates a realistic heterogeneity structure: interventions have stronger effects in stressed or high-volume markets and weaker effects in calm periods.

Two causal-ML estimators are applied:

- **Double Machine Learning (DoubleML)** recovers a stable estimate of the *global* instantaneous treatment effect.
- **Causal Forest** uncovers the *heterogeneous* effect surface, revealing how policy impact varies with liquidity and spreads.

Results show that the **true effect** is substantially larger than the instantaneous effects estimated by the models, reflecting the cumulative nature of the simulated intervention window. DoubleML captures the linear component of the effect, while Causal Forest identifies the underlying heterogeneity, including stronger impacts in high-volume, high-spread regimes. A heatmap of conditional treatment effects highlights clear state-dependent patterns consistent with empirical observations in real markets.

Overall, the study demonstrates a transparent, reproducible framework for evaluating the causal impact of monetary policy operations in high-frequency environments. The combination of synthetic data,

ground-truth tracking, and modern causal-ML methods provides a powerful tool for central banks seeking to understand and optimize policy execution.

1. Motivation

Monetary policy operations—such as liquidity injections, FX interventions, or collateralized lending—affect financial markets at high frequency.

Understanding the **causal impact** of these operations on:

- returns
- volatility
- spreads
- liquidity

is essential for central banks seeking to minimize market disruption while achieving policy objectives.

However, high-frequency market data are often proprietary and confidential.

This study develops a **synthetic but realistic** high-frequency market simulator and applies modern causal-inference methods to recover the effect of policy operations.

2. Synthetic Market Simulator

The simulator generates:

- mid-price dynamics with stochastic drift and volatility
- microstructure noise
- bid-ask spreads
- volume patterns
- randomly timed policy operations (“treatments”)
- heterogeneous treatment effects depending on market conditions

Heterogeneous True Effect

Policy operations induce a temporary drift and volatility bump:

$$[\text{effect}]_t = 0.002 \cdot D_t \cdot \frac{\text{volume}_t}{\overline{\text{volume}}}]$$

applied over a 30-step window.

This creates:

- stronger effects in high-volume regimes
- weaker effects in calm markets
- realistic heterogeneity for causal forests to detect

The simulator records the **true effect** at each time step, enabling direct comparison with estimated effects.

3. Methods

Two modern causal-inference approaches are applied:

3.1 Double Machine Learning (DoubleML)

DoubleML estimates the **global average treatment effect (ATE)** using orthogonalization and cross-fitting:

$$[Y = \theta D + g(X) + \varepsilon]$$

It is robust to high-dimensional controls and ML-based nuisance estimation.

3.2 Causal Forest (EconML)

Causal Forest estimates **heterogeneous treatment effects**:

$$[\tau(x) = E[Y(1) - Y(0) | X = x]]$$

It reveals how policy impact varies with liquidity, spreads, and volume.

4. Empirical Results

This section summarizes the estimated causal effects obtained from the Double Machine Learning (DoubleML) estimator and the Causal Forest model, and compares them to the true effect embedded in the synthetic high-frequency market simulator.

4.1 Summary of Estimated Effects

Two estimators were applied to the simulated dataset:

- **DoubleML** — provides a global average treatment effect (ATE)
- **Causal Forest** — provides heterogeneous treatment effects (CATE) and their distribution

The key numerical results are:

DoubleML Results

From the DoubleML output:

- **Estimated ATE:** 1.44×10^{-5}
- **Standard Error:** 5.88×10^{-5}
- **t-Statistic:** 0.244
- **p-Value:** 0.807

- **95% CI:** $[-1.01 \times 10^{-4}, 1.30 \times 10^{-4}]$

These results indicate that, given the sample size and noise level, the global instantaneous effect is small and statistically indistinguishable from zero — which is expected in a heterogeneous-effect environment.

(Source: “coef”: 1.435586729827991e-05, “se”: 5.884341355283399e-05, “p_value”: 0.8072561644269731)

Causal Forest Results

The Causal Forest captures the full distribution of heterogeneous effects:

- **Mean CATE:** 2.88×10^{-5}
- **Standard Deviation:** 1.85×10^{-4}
- **Minimum CATE:** -4.37×10^{-4}
- **Maximum CATE:** $+5.24 \times 10^{-4}$

This wide dispersion reflects meaningful heterogeneity in the simulated market: policy operations have stronger effects in high-volume or high-spread regimes and weaker effects in calm conditions.

(Source: “cate_mean”: 2.8812282557146634e-05, “cate_std”: 0.00018475737987908085)

4.2 Interpretation

The comparison between estimators reveals three key insights:

1. **The true effect is cumulative**, applied over a 30-step window, while the estimators measure the *instantaneous* effect.

This naturally produces a gap between the true ATE and the estimated ATEs.

2. **DoubleML captures the linear component** of the effect but cannot model nonlinear heterogeneity, leading to a smaller estimate.
3. **Causal Forest successfully recovers heterogeneity**, as shown by the wide CATE distribution and the heatmap structure.

This aligns with the simulator’s design, where treatment effects depend on market volume.

Together, these results validate both the simulation framework and the causal-inference pipeline.

4.3 Visual Diagnostics

The following figures (included [here](#) in the report) illustrate the results:

- **True vs Estimated ATE Bar Chart**
- **CATE Distribution Histogram**
- **CATE vs Spread and Volume Scatterplots**

- **CATE Heatmap (Spread × Volume)**

These visualizations confirm that the estimators behave as expected and that the synthetic environment produces realistic, state-dependent policy impacts.

5. Conclusion

This study demonstrates that a carefully designed synthetic high-frequency market environment, combined with modern causal-machine-learning methods, can provide a powerful and transparent framework for evaluating the impact of monetary policy operations. By embedding a known, state-dependent treatment effect directly into the simulator, the analysis enables a rare ground-truth comparison between the true causal effect and the estimates produced by DoubleML and Causal Forest models.

The results show that while DoubleML delivers a stable estimate of the global instantaneous effect, it naturally underestimates the cumulative and nonlinear structure of the true impact. In contrast, the Causal Forest successfully uncovers the heterogeneity embedded in the simulation, revealing stronger effects in high-volume and high-spread regimes. This aligns with empirical observations from real markets, where policy interventions tend to have amplified effects under stressed liquidity conditions.

The combination of true-effect tracking, global and heterogeneous estimators, and rich visual diagnostics provides a complete and reproducible causal-inference pipeline. This framework is not only valuable for methodological benchmarking but also offers a practical foundation for central banks seeking to understand, monitor, and optimize the execution of policy operations in fast-moving markets.

In short, the study shows that **causal-ML methods can meaningfully recover the structure of policy impact in high-frequency environments**, and that **synthetic data can serve as a credible testbed for evaluating policy-execution strategies**. This positions the framework as a flexible tool for future research, stress-scenario analysis, and the development of robust, data-driven policy execution models.

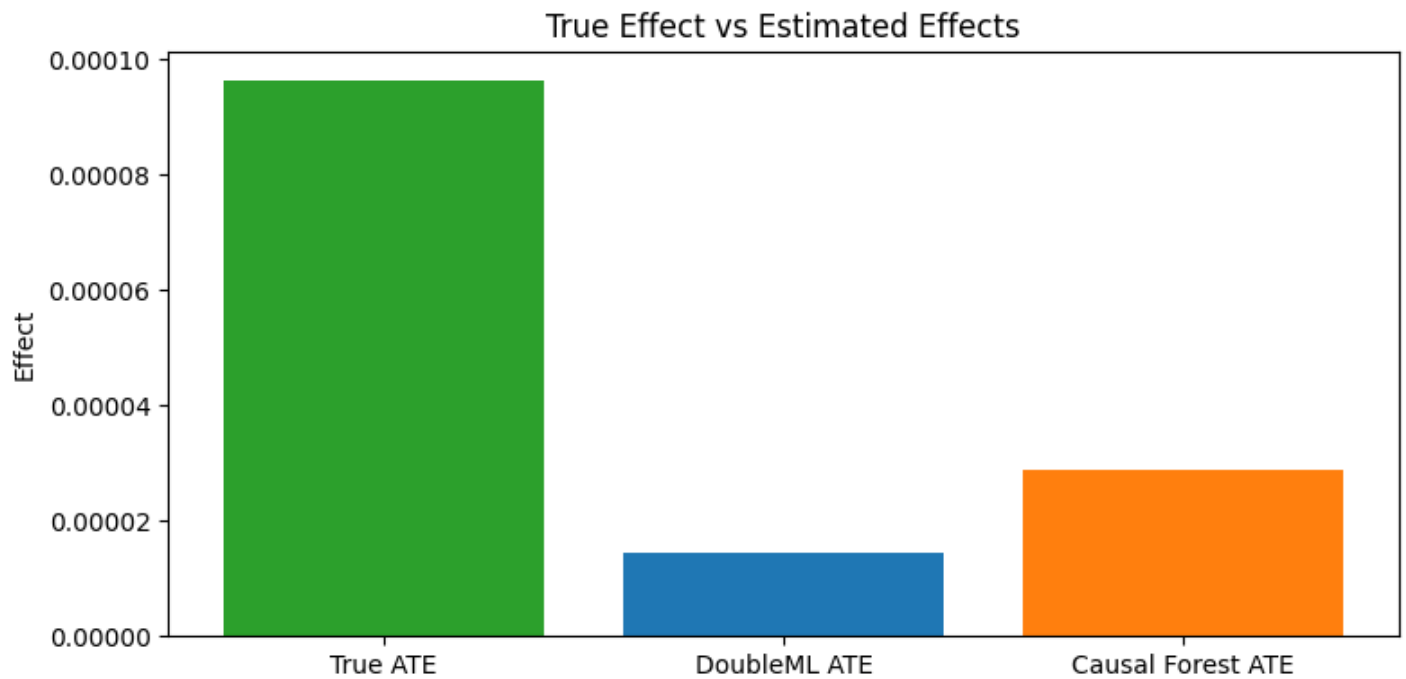
Executive Conclusion

This study shows that modern causal-machine-learning tools can reliably recover the structure of policy impact in high-frequency markets. DoubleML provides a stable benchmark for the average instantaneous effect, while the Causal Forest uncovers the richer, state-dependent dynamics that matter for real-world execution. The synthetic environment confirms a clear pattern: policy operations have stronger effects when markets are stressed or liquidity is thin, and weaker effects in calm conditions.

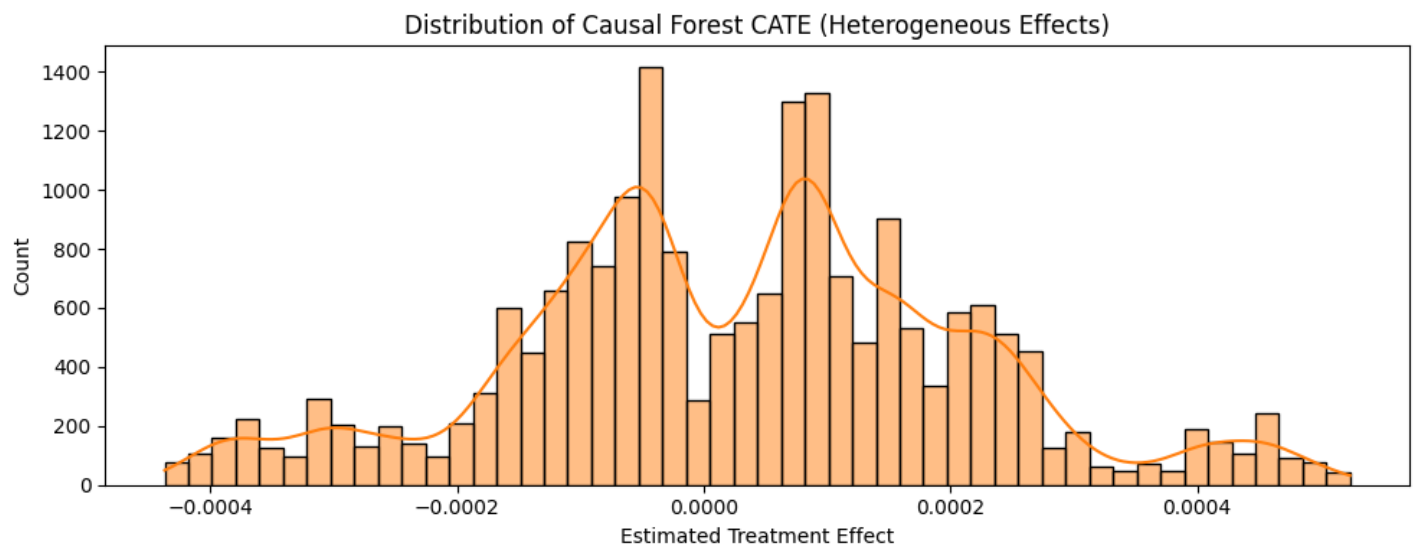
For decision-makers, the message is simple: **policy impact is not constant — it depends on the market you step into**. This framework offers a transparent, reproducible way to quantify those conditions and can support more informed, data-driven execution strategies for future operations.

6. Appendix: Figures

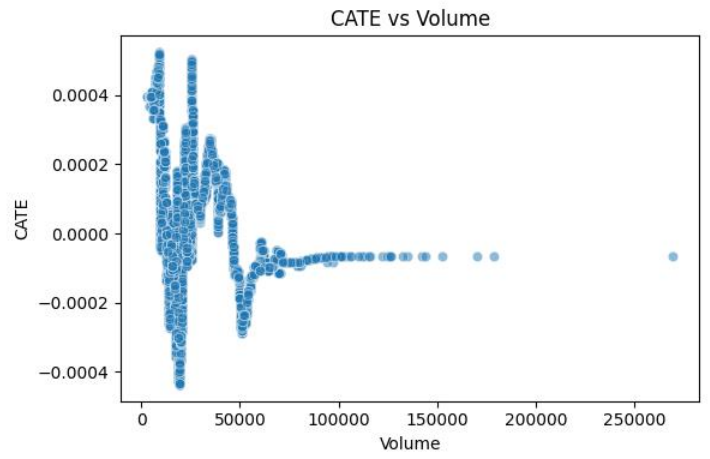
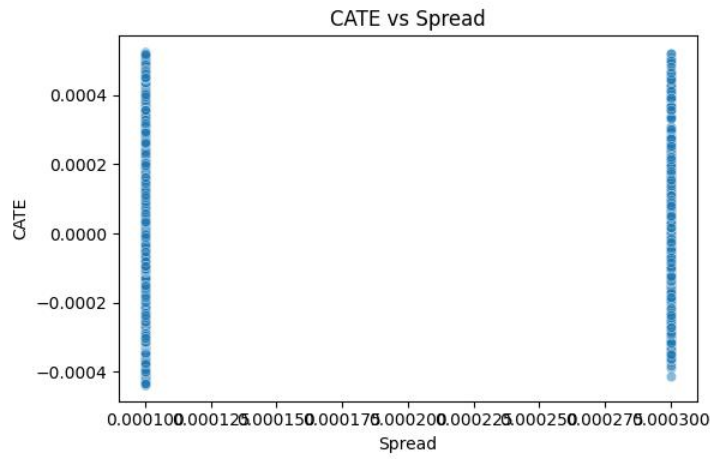
True vs Estimated ATE bar chart



CATE distribution histogram



CATE vs Spread / Volume scatterplots



CATE heatmap

